When Image and Text Disagree: Cross-Modal Evidence Conflict in Multimodal Retrieval-Augmented Generation

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We introduce **CMC-Bench** — the Cross-Modal Conflict Benchmark — to evaluate multimodal RAG systems under contradicting image/text evidence. Across **3,768 instances** drawn from ChartQA and MMMU, four open VLMs are tested under four conflict conditions and four conflict types. All models exhibit a *modality lean* rather than reliable evidential arbitration (MFR 0.430–0.486); a single hallucination rate obscures whether a model is **abstaining appropriately** or **fabricating**; and cross-modal conflict degrades accuracy by ΔAcc 0.17–0.46.

RESULTS

Model	Aligned	Image-correct	Text-correct	Both-wrong	MFR	ConfabR	CDR	ΔAcc
Qwen3-VL-4B	.898	.409	.375	.675	.486	.189	**.317**	.412
LFM2.5-VL-1.6B	**.925**	.235	**.870**	.277	.461	.137	.050	.464
Gemma-3n-E2B	.599	.275	.193	**.822**	.430	.519	.187	**.169**
DeepSeek-VL2-s	.901	.334	.843	.279	.485	.132	.081	.416

Orange = best per column. **Grey** = worst. n = 942 per condition. Gemma's best ΔAcc (.169) co-occurs with highest ConfabR (.519) — artefact of fabrication, not robustness. Qwen3's CDR (.317) reflects principled conflict detection, not failure.

1 MOTIVATION & PROBLEM

Multimodal RAG systems retrieve and fuse *both* images and text to answer questions. A critical but understudied failure mode occurs when the two modalities **contradict each other** — e.g. a chart reporting 64% while the retrieved passage states 41%, or an image dated 2023 while the text describes 2021.

Under cross-modal conflict, correct generation requires **arbitration** (select the evidentially correct source) or responsible **abstention** (flag the contradiction when neither source is reliable).

Research gap: No benchmark had systematically evaluated VLMs when retrieved image and text evidence directly contradict, using realistic multi-modal RAG conditions.

We ask four questions: **RQ1** Does conflict degrade accuracy? **RQ2** Do models arbitrate by evidence or follow fixed priors? **RQ3** Which modality exerts stronger evidential pull? **RQ4** Does a single HR metric adequately capture hallucination?

2 CONFLICT TAXONOMY

- **Factual Contradiction**
  Image value A vs. text value B — same question, different numbers.
  e.g. chart: 64% · passage: "41%"
- **Temporal Mismatch**
  Image depicts period X; text describes period Y.
  e.g. chart title "2023" · passage: "In 2021..."
- **Entity Confusion**
  Image depicts entity A; text describes a visually/semantically similar entity B.
  e.g. similar-looking geographic regions, related species
- **Granularity Conflict**
  Image is a specific instance; text states a general rule — or vice versa.
  e.g. chart: one country · text: "Across all regions..."

3 BEHAVIORAL LABEL SPACE

Label	Behavior	Correct under
IMAGE	Aligns with image source	Img-correct
TEXT	Aligns with text source	Txt-correct
BOTH	Aligns with both sources	Aligned
NEITHER	Fabricated — aligns with neither	—
ABSTAIN	Refused / flagged conflict	Both-wrong

HR = ConfabR (NEITHER rate) + CDR (ABSTAIN rate). Merging them into one metric conflates opposite behaviors — see §⑦.

4 BENCHMARK CONSTRUCTION

1. **Source data:** ChartQA eval split (factual & temporal conflicts) and MMMU eval split (entity & granularity conflicts). Eval splits only — no contamination with training data.
2. **Passage synthesis:** Aligned passages state the gold answer. Contradicting passages use the same template but substitute a *same-type* wrong answer drawn from the pool.
3. **Wrong-image sampling:** ChartQA — sampled within the same conflict-type pool. MMMU — from the same subfield when available, otherwise the same domain.
4. **Four conditions** per base example: Aligned · Img-correct · Txt-correct · Both-wrong
5. **LLM-as-judge** (GPT-4o-mini) classifies each VLM response into the five-way label space without access to the gold answer. 20% randomly sampled for manual quality review (κ = 0.81).

5 SCALE

942	3,768	4	4	5
BASE EXAMPLES	TOTAL INSTANCES	CONFLICT TYPES	OPEN VLMS	BEHAVIORAL LABELS

7 QUALITATIVE EXAMPLE

Q: "Which year had the highest native-born employment in Slovenia?"
Image gold: 2008
Text passage: "...the value is 2015"

Model	Response (excerpt)	Label	Fair?
DeepSeek	"2008"	IMAGE	✓
LFM2.5	"…the value is 2015."	TEXT	✗
Gemma	"…employment in 2004."	NEITHER	✗
Qwen3	"Chart covers 2004–2012; text cites 2015 which lies outside that range…"	ABSTAIN	✓

8 KEY FINDINGS

RQ1 **Conflict severely degrades accuracy.** ΔAcc 0.17–0.46 across all models. The smallest ΔAcc (Gemma, 0.169) is an artefact: high ConfabR (.519) means the model fabricates rather than committing to a wrong modality, artificially shrinking the gap.

RQ2 **No model reliably follows the evidentially correct modality.** MFR 0.430–0.486 — a narrow 5.6-pt spread across architecturally diverse models, consistent with fixed modality priors rather than instance-level evidential arbitration.

RQ3 Text exerts stronger **evidential pull under conflict** than image — defined as correctness-conditional accuracy when the modality is right. LFM2.5 *Txt-C .870 vs Img-C .235* (Δ = 0.635). Paradoxically, MPB shows a slight image-label majority in unconstrained conditions, suggesting models default toward image outputs when no modality is definitively correct — but follow text far more reliably when it is.

RQ4 **A single HR conflates opposite behaviors.** Qwen3's HR = 0.507 is dominated by CDR (.317): it detects irresolvable conflict. Gemma's HR = 0.706 is dominated by ConfabR (.519): it fabricates. *CDR should be treated as a **positive** indicator.*

9 HR DECOMPOSITION

Qwen3-VL-4B
ConfabR (NEITHER)
CDR (ABSTAIN)
18.9% 31.7% ✓

LFM2.5-VL-1.6B
ConfabR (NEITHER)
CDR (ABSTAIN)
13.7% 5.0%

Gemma-3n-E2B ⚠
ConfabR (NEITHER)
CDR (ABSTAIN)
51.9% 18.7%

DeepSeek-VL2-s
ConfabR (NEITHER)
CDR (ABSTAIN)
13.2% 8.1%

10 MODALITY PREFERENCE (MPB)

Qwen3-VL-4B
IMG 55%
Txt-C **.375** · Img-C **.409** Δ -0.03

LFM2.5-VL-1.6B
TXT 53%
Txt-C **.870** · Img-C **.235** Δ +0.64

Gemma-3n-E2B
IMG 57%
Txt-C **.193** · Img-C **.275** Δ -0.08

DeepSeek-VL2-s
IMG 52%
Txt-C **.843** · Img-C **.334** Δ +0.51

11 CONCLUSION

CMC-Bench reveals that open VLMs default to **static modality priors** under cross-modal conflict rather than per-instance evidential arbitration. Evaluations must decompose HR into ConfabR and CDR — penalising abstention equally with fabrication is an *evaluation design flaw*. Future multimodal RAG systems must explicitly support and reward **conflict-aware generation**.

github.com/jaspercatapang/cmc-bench
3,768 instances · prompts · judge spec · code — all public
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Models evaluated via mlx-vlm (4-bit, Apple Silicon M3 Pro) · Supported by Money Forward Inc. research funding · ACL 2026 Workshop Submission

Qwen3-VL-4B · LFM2.5-VL-1.6B
Gemma-3n-E2B · DeepSeek-VL2-small